Applying Simple Intertemporal Economics to Covid-19

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Introduction

- What use is intertemporal thinking?
- Simple but important application: how willing should we be to trade off lives for economic harm in COVID-19 pandemic?
- If only killed 500,000 terminally-ill people with a few minutes left to live, probably not worth \$3 trillion for 60 minutes more each (\$100,000/life-minute).
- But if save lives of 500,000 people who would otherwise live for 60 more years, potentially worth it (\$0.19 per life-minute saved)
- IVT-type argument: there's a place where we're indifferent!
- Worth stopping here: do you agree? Because many politicians and some economists claim not to agree.

Hall, Jones, and Klenow (2020)

- Enter Hall, Jones, and Klenow (2020)
- How do we think about that tradeoff?
- Write down a simple model of people consuming and risking death. So, value V of living at some age a is, given probability of given age a at time t of $S(a, t)$:

$$V_a = \sum_{t=0}^{\infty} \beta^t S(a, t) u(c_t)$$

  - V_a: value of being alive at age a
  - $\sum_{t=0}^{\infty}$: sum over all future periods
  - β^t: discount the future
  - $S(a, t)$: only value if you survive
  - $u(c_t)$: get utility from consumption/being alive
- Do you disagree? If so, let us know in discussion forum!

Value of being alive

- Value of being alive at age a is:

$$V_a = \sum_{t=0}^{\infty} \beta^t S(a,t) u(c_t)$$

- Can rewrite this as

$$V_a = u(c_0) + \beta S(a,1) V_{a+1}$$

- In other words, value of being alive at age a is initial utility plus discounted (β) value of waking (V_{a+1}) up tomorrow, times probability you actually wake up ($S(a,1)$)

Social Planner (Politician's) Problem

- What is trade off problem for benevolent social planner/politician?
- Need to know populations: N is number of people (normalize old population to 1)
- COVID-19 reduces survival rate of old to $S - \delta$ for one period
- What is tradeoff between reduction in consumption λ and δ? (denote young y and old o)

$$W(\lambda, \delta) = (N - 1)V_y + 1 \cdot V_0$$

- Utilitarian treating people equally here

Maximizing Welfare

- The social planner wants to choose consumption drop λ and death rate reductionδ to maximize utility:

$$\begin{aligned} W(\lambda, \delta) &= (N-1)V_y + 1 \cdot V_0 \\ &= (N-1)\left[u(\lambda c_0) + \beta S_y V(F, y)\right] + \left[u(\lambda c_0) + \beta(S_0 - \delta)V_F\right. \end{aligned}$$

- Where V_F is utility if survive
- $W(\lambda, 0)$ is welfare when death rate is zero (but consumption reduced by some amount)
- $W(1, \delta)$ is welfare when consumption does not decline but death rate is untouched (baseline)
- How much are we willing to pay to reduce death rate to zero? λ that solves this problem tells us: $W(\lambda, 0) = W(1, \delta)$

Skipping Math

- The authors take the equation, take a Taylor expansion of utility around no drop in consumption (linearize utility functions(!)) and plug into our problem to find the tradeoff:

$$1 - \lambda = \frac{\beta\delta}{N} \frac{V_{F,o}}{u'(c_0)c_0}$$

- Letting $\alpha = 1 - \lambda$ be the amount you'd give up to solve the crisis completely, and letting LE be life expectancy of old:
- Also let $v = u(c)/(u'(c)c)$ be value of one more year of life relative to consumption (convert consumption to life-years)

$$\alpha = \frac{\delta v LE_o}{N}$$

- α: Pay in fraction of consumption drop
- δ: Pay more when death rate higher
- v: Pay more when life more than consumption
- LE_o: Pay more when old have long life ahead of them
- N: Pay less when they are a smaller fraction of population

Calibrate the Model

- The following are the author's best-guesses from the literature:
  - Death rate of 0.81% without mitigation efforts $\delta = 0.0081$
  - Consumption represents 1/3 of happiness in life $v = 3$
  - Life expectancy $LE_o = 14.47$ years
  - $N = 1$ (first assume everyone is facing the same mortality risk)
- From these, get $\alpha = 0.35$! Willing to pay 35% of consumption to solve crisis.

Robustness

Table 1: Willing to Give Up What Percent of Consumption? (Representative Agent)

Mortality rate, δ (percent)	— Value of Life, v — 2	3	4
Using Taylor series linearization:			
0.41	11.7	17.6	23.4
0.81	23.4	35.2	46.9
1.62	46.9	70.3	93.8
Using CRRA utility with $\gamma = 2$:			
0.41	10.5	15.0	19.0
0.81	19.0	26.0	31.9
1.62	31.9	41.3	48.4

Calibrate the Model

- Assumption that everyone is young is dumb!
  - Death rate of 3.86% without mitigation efforts for old, including virus contraction rate without mitigation of 75% (is this a good assumption?)
  - Life expectancy shorter: $LE_o = 10.9$
- From these, get $\alpha = 0.21$! Willing to pay 21% of consumption to solve crisis.
- Note that if current mitigation only drops 50%, willing to pay only 10.5% drop

Robustness

Mortality rate, δ (percent)	— Value of Life, v — 2	3	4
Using Taylor series linearization:			
2	7.3	10.9	14.5
3.86	14.0	21.0	28.0
5	18.2	27.2	36.3
Using CRRA utility with $\gamma = 2$:			
2	6.8	9.8	12.7
3.86	12.3	17.4	21.9
5	15.4	21.4	26.7

Iconoclastic Calibration

- Death rate of older of 5.15%, but let's say due to asymptomatic/non-reporting of those who don't have complications it's really 70% that. Assume older mitigation is 60% getting it
- But drop in death rate is only half (we aren't perfect): $\delta = 0.0515 * 0.7 * 0.6 * 0.5 = 0.0108$
- For "realistic" mitigation efforts, should only be willing to pay:

$$\alpha = \frac{0.0108 \cdot 3 \cdot 10.9}{5} = 7.06\%$$

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Comparing other Estimates

- These give willingness to pay of between \$1.3 trillion and \$6 trillion
- Luigi Zingales: \$9 million per lost life above 65, 7.2 million deaths prevented, \$65 trillion
- Why so different? Way higher (6x) on value of life and death rate (4.5x) (9 million baseline deaths compared to 2 million in Hall, Jones, and Klenow)
- Fortunately these are empirical questions

Conclusions/Discussion

- People make tradeoffs between life expectancy and consumption all the time
- Crazy (welfare-decreasing) to make dramatically different decisions than they would make
- People are willing to trade of life-minutes for own consumption (and consumption of their children)
- My own conclusions: **on utilitarian grounds** honest people can disagree whether or not a decline between 7% and 30% of consumption is a reasonable cost to pay, depending on how effective mitigation and on facts
- Outside those bounds, probably making the average person worse off (but maybe you have moral reasons, or a calibration like Zingales's)
- Benefit of economic thinking: you don't have to call people ghouls or death eaters when they disagree–just correct their calibration or model!